1 Explanation certificate: churn100-s4-c2

This section is the explanation certificate of the churn100-s4-c2 network. It is rendered from the canonical JSON record `churn100-s4-c2.json` (schema 1.4.0, SHA-256 `36f617b704b0542d...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s4, seed 4
Genome	`4206744f-b5ce-4488-bc53-a39aa08fe972` (NEAT key 24427, generation 99)
Explanation	`33d6db5d-fbdf-41c8-a597-fc997715b92f` (88 operations)
Train AUC / accuracy	0.8908 / 0.888 (n = 4000)
Test AUC / accuracy	0.8664 / 0.898 (n = 1000)
Output activation	sigmoid
Phenotype	85 nodes (71 inputs, 13 hidden, 1 output), 58 enabled edges
Explained model state	93 nodes, 60 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 1, split_node 5
Function preservation	max $\|\Delta \text{output}\| = 4.44 \times 10^{-16}$ on the training split
Annotations	9 (7 leaf, 2 composition)
Structural coverage	1 (92 of 92 candidate nodes; 37 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`36f617b704b0542d...`

Table 1: Explanation certificate summary: churn100-s4-c2.

Drafter's summary. logit P(churn) = -1.30 + a sparse linear backbone over 29 inputs (voicemail plan -1.12 per SD, international plan +0.83, voicemail messages +0.62, evening charge +0.35, international minutes +0.34, state offsets) + a customer-service-calls threshold (two hinges: nothing for 0-2 calls, then about +2.4 logit per call from 4) + two day-usage hinges (nothing below about 210 day minutes, then steeply up, +4.3 logit above 280 minutes; the 3118 hinge only for customers without voicemail messages, the 10131 hinge off in state:47) + a small evening-minutes hinge and exact state:39/state:15 offsets, all additive in the logit. Day minutes and service calls have no linear term. Train AUC 0.891, accuracy 0.888. Compared with the run's cluster-1 lead (churn100-s4, key 24573), the same two thresholds (service calls, high day usage) carry the nonlinearity, but here day usage is gated by voicemail messages and state:47 rather than combined with evening minutes.

What resisted explanation. Nothing structural: no hidden node needed splitting (5 input splits, 1 consolidation of the customer-service-calls copies). 6089 (state:18 / 21+ voicemail messages) has logit weight 0.005 (at most +0.05): negligible. The state:39 pair (1295 -> 7872) is a constant -2.33 logit switched off for state:39, almost cancelling state:39's direct -2.61 (net -0.25 vs empirical -0.93). The service-call threshold is placed right (3 -> 4 calls) but the model under-predicts at 4 calls (0.33 vs 0.44) and is over-confident from 5 calls (0.75-0.99 vs 0.55-0.68; 119 customers). state:18 gets +0.41 logit against an empirical -0.48 (65 rows). closed_form_max_abs_err 0.020 (vmail_state18_unit) is 3-significant-figure rounding; numeric error 1.8e-15.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 3%; it adds up six parts: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] A higher evening charge and more international minutes also raise it steadily, each from about 2% to about 4% across their usual range. [C055, C056] Through it, the state moves the chance from about 3% in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] This part adds up two parts: a rise from 3 customer-service calls upwards and a second rise from 4 calls upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] This part reads only the state: it raises a typical customer's chance from about 3% to about 22% in Rhode Island, and by less than a point in Indiana. [C100, C101] This part reads evening minutes: nothing until about 217 minutes, then steadily, but a typical customer's chance stays at about 3%. [C103, C104] This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions (by at most one point). [C111]

The whole model. The model adds up six parts into one score: fixed-weight inputs, service calls, heavy daytime use, state units, an evening unit and a small voice-mail unit. [C046] *Evidence.* It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C046] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 3% to about 31%, and a voice-mail plan lowers it to under 1%. [C001, C006] More daytime minutes raise it in an uneven way, from about 3% at 112 minutes to about 8% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C004] In Washington the chance is about 11%, against about 3% in West Virginia. [C002] The model ignores evening calls, international calls, the international charge, night-time calls, the night-time charge and night-time minutes. [C040, C041, C042, C043, C044, C045]

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan, voice-mail messages, the evening charge and international minutes raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C047, C049, C055, C056, C051] *Evidence.* Through it, an international plan takes a typical customer from about 3% to about 31%, and a voice-mail plan lowers the chance to under 1%. [C047, C051] More voice-mail messages raise it steadily, from about 3% at 0 to about 10% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C049] Through it, the state moves the chance from about 3%

in West Virginia to about 11% in Washington and about 7% in New Jersey, and to under 1% in Hawaii and in Rhode Island (72 to 122 training customers per state). [C048, C050, C052, C053] Its formula reproduces the part exactly on all 4000 training customers. [C076]

Service calls. This part raises the chance of leaving with calls to customer service from 3 calls upwards, slowly at first, then faster; it adds up two parts. [C079, C078, C077] *Evidence.* It is made of two parts that are added: calls from 3 upwards and calls from 4 upwards. [C079] From 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 3% to about 18%. [C078] Up to 3 calls it barely moves a typical customer's chance, which stays at about 3%, rising slowly at first, then faster. [C077]

Calls from 4 upwards. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C093] *Evidence.* It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 8% without it and about 18% with it. [C094] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 6%. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C095]

Calls from 3 upwards. This part raises the chance of leaving from 3 customer-service calls upwards, slowly at first, then faster. [C097, C096] *Evidence.* It acts on 848 of 4000 training customers, 215 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 4% with it. [C098] Through it, the chance rises slowly at first, then faster: 4 calls take a typical such customer from about 3% to about 8%. [C097] Its formula reproduces the part exactly on all 4000 training customers. [C099]

Heavy daytime use. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way, and voice-mail messages and being in Washington lower it. [C088, C080, C082] *Evidence.* This part acts on 1099 of 4000 training customers, 291 of whom left; for the typical customer it acts on, the chance is about 3% without it and about 12% with it. [C091] Daytime minutes add to the chance only once they rise above about 221 minutes (between 186 and 221 minutes for different customers), in an uneven way: for a typical such customer, from about 3% at 0 minutes to about 19% at 271 minutes. [C088] More voice-mail messages lower it in an uneven way: for the typical customer it acts on, from about 12% at 0 to about 6% at 32 messages. [C080] Customers in Washington get less from it: about 7% instead of about 12% for the typical customer it acts on. [C082] Its formula reproduces the part exactly on all 4000 training customers. [C092]

State units. This part reads only the state and raises the chance of leaving in Rhode Island, and a little in Indiana. [C100, C101] *Evidence.* Through it, a typical customer's chance goes from about 3% to about 22% in Rhode Island, and up by less than a point in Indiana. [C100, C101] Its formula reproduces the part exactly on all 4000 training customers. [C102]

Evening unit. This part reads evening minutes and barely changes the chance of leaving: nothing until about 217 minutes, then steadily. [C103, C104] *Evidence.* Through it, evening minutes change nothing until about 217 minutes, then raise the chance steadily, but a typical customer's chance stays at about 3% from 138 to 266 minutes. [C103] Its formula reproduces the part exactly on all 4000 training customers. [C105]

Small voice-mail unit. This small part reads voice-mail messages and being in Louisiana, and barely changes the predictions. [C111] *Evidence.* Through it, a typical customer's chance stays at about 3% in Louisiana as elsewhere, and voice-mail messages barely change it (about 3% at 0 and

at 32 messages). [C106, C107] Switching it off changes the predictions by at most one point. [C111] Its formula reproduces the part exactly on all 4000 training customers. [C112]

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to under 1% while the state units raise it to about 22%, so overall it stays at about 3%. [C053, C100, C033] Worth checking: the state moves the chance by several points (Washington about 11%, New Jersey about 7%, Hawaii under 1%), based on 72 to 122 training customers per state. [C002, C005, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C006, C003] Worth checking: voice-mail messages raise the chance through the fixed-weight inputs but lower it through heavy daytime use. [C049, C080]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7; range across patients 2.82 to 2.82	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C002	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.27, OR 3.58; range across patients 0.64 to 1.48	122, 72	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	on the training data only
C003	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	log-odds 1.21, OR 3.35; range across patients 0.39 to 1.47	4000	train; test agrees (AUC gain 0.017 [0.005, 0.031], n=1000)	yes	clearly
C004	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	log-odds 1.13, OR 3.10; range across patients 0.83 to 1.79	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C005	direction: state: West Virginia to New Jersey (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 7%	log-odds 1.09, OR 2.97; range across patients 1.09 to 1.09	122, 90	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C006	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08; range across patients -2.54 to -2.54	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C007	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	122, 73	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C008	direction: state: West Virginia to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.67, OR 1.95; range across patients 0.67 to 0.67	122, 65	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C009	direction: state: West Virginia to Connecticut (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.66, OR 1.94; range across patients 0.66 to 0.66	122, 80	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C010	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.88, OR 2.42; range across patients 0.88 to 0.88	4000	train; test agrees (AUC gain 0.013 [0.004, 0.022], n=1000)	yes	slightly
C011	direction: total international minutes: 7 minutes to 14 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.85, OR 2.35; range across patients 0.85 to 0.85	4000	train; test unclear (AUC gain 0.0008 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C012	direction: state: West Virginia to Vermont (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to <1%	log-odds -1.68, OR 0.19; range across patients -1.68 to -1.68	122, 80	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	122, 68	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C014	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.57, OR 1.77; range across patients 0.57 to 0.57	122, 86	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C015	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 84	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	122, 82	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C017	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.47, OR 1.61; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C018	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 91	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C019	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	122, 66	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 79	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C021	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.33, OR 1.40; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.0008], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Georgia (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40	122, 60	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.37, OR 0.69; range across patients -0.37 to -0.37	122, 56	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 68	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: state: West Virginia to state other than Georgia, Hawaii, Indiana, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 2286	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C026	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C027	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.22, OR 1.25; range across patients 0.22 to 0.22	122, 73	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C028	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: hinge at 218 minutes; 100% of patients same; verified; risk saturates near 0%; matches the fold at 217 minutes*	3% to 3%	log-odds 0.20, OR 1.22; range across patients 0.20 to 0.20	4000	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	4000	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	slightly
C030	direction: state: West Virginia to Arkansas (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.14, OR 1.15; range across patients 0.14 to 0.14	122, 78	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C031	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0004, 5e-05], n=1000)	yes	slightly; on the training data only
C032	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (whole model) *shape: hinge at 42.20 US dollars; 77% of patients same; fallback; risk saturates near 0%; matches the fold at 39.34 US dollars*	3% to 3%	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.94	4000	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; on the training data only
C033	direction: state: West Virginia to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.004, OR 1.00; range across patients 0.004 to 0.004	122, 82	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Florida (whole model) *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 3%	log-odds -0.0006, OR 1.00; range across patients -0.0006 to -0.0006	122, 78	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	slightly; on the training data only
C035	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C036	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, New Jersey, Louisiana, Connecticut, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.48, OR 4.40	122, 72	train; test disagrees (AUC gain -0.012 [-0.026, 0.0005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 72% of patients same; fallback; risk saturates near 0%*	3% to 11%	log-odds 1.47, OR 4.37	4000	train; test agrees (AUC gain 0.017 [0.005, 0.031], n=1000)	yes	clearly
C038	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: hinge at 223 minutes; 50% of patients same; fallback; risk saturates near 0%; matches the fold at 221 minutes*	3% to 8%	log-odds 1.15, OR 3.17	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C039	magnitude: voice-mail plan: no voice-mail plan to a voice-mail plan, typical patient *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C040	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C041	null: model ignores total international calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C042	null: model ignores total international charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C043	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C044	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C045	null: model ignores total night-time minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C046	structure: made of linear_backbone, service_calls, day_usage_hinges, evening_hinge, vmail_state18_unit, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C047	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	3% to 31%	log-odds 2.82, OR 16.7; range across patients 2.82 to 2.82	3620, 380	train; test agrees (AUC gain 0.060 [0.032, 0.090], n=1000)	yes	clearly
C048	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 11%	log-odds 1.48, OR 4.40; range across patients 1.48 to 1.48	122, 72	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	3% to 10%	log-odds 1.47, OR 4.35; range across patients 1.47 to 1.47	4000	train; test agrees (AUC gain 0.016 [0.003, 0.029], n=1000)	yes	clearly
C050	direction: state: West Virginia to New Jersey (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 7%	log-odds 1.09, OR 2.97; range across patients 1.09 to 1.09	122, 90	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C051	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	3% to <1%	log-odds -2.54, OR 0.08; range across patients -2.54 to -2.54	2939, 1061	train; test agrees (AUC gain 0.033 [0.014, 0.053], n=1000)	yes	slightly
C052	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.50, OR 0.08; range across patients -2.50 to -2.50	122, 73	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C053	direction: state: West Virginia to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -2.36, OR 0.09; range across patients -2.36 to -2.36	122, 82	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C054	direction: state: West Virginia to Connecticut (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.66, OR 1.94; range across patients 0.66 to 0.66	122, 80	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C055	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.88, OR 2.42; range across patients 0.88 to 0.88	4000	train; test agrees (AUC gain 0.013 [0.004, 0.022], n=1000)	yes	slightly
C056	direction: total international minutes: 7 minutes to 14 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 4%	log-odds 0.85, OR 2.35; range across patients 0.85 to 0.85	4000	train; test unclear (AUC gain 0.0008 [-0.007, 0.008], n=1000)	yes	slightly; on the training data only
C057	direction: state: West Virginia to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.63, OR 1.87; range across patients 0.63 to 0.63	122, 65	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C058	direction: state: West Virginia to Vermont (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to <1%	log-odds -1.68, OR 0.19; range across patients -1.68 to -1.68	122, 80	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C059	direction: state: West Virginia to Tennessee (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.60, OR 1.82; range across patients 0.60 to 0.60	122, 68	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C060	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 5%	log-odds 0.57, OR 1.77; range across patients 0.57 to 0.57	122, 86	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 84	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C062	direction: state: West Virginia to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.47, OR 1.61; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C063	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 1%	log-odds -0.69, OR 0.50; range across patients -0.69 to -0.69	122, 91	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C064	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.32, OR 1.37; range across patients 0.32 to 0.32	122, 66	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C065	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 4%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 79	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C066	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.33, OR 1.40; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.002 [-0.005, 0.0008], n=1000)	yes	slightly; on the training data only
C067	direction: state: West Virginia to Georgia (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.40, OR 0.67; range across patients -0.40 to -0.40	122, 60	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C068	direction: state: West Virginia to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 2%	log-odds -0.37, OR 0.69; range across patients -0.37 to -0.37	122, 56	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.30; range across patients 0.26 to 0.26	122, 68	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C070	direction: state: West Virginia to state other than Georgia, Hawaii, Kentucky, Louisiana, Massachusetts, Arkansas, North Dakota, Arizona, New Jersey, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Dakota, Tennessee, Virginia, Vermont, Washington, West Virginia, Connecticut, Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.26, OR 1.29; range across patients 0.26 to 0.26	122, 2368	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C071	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	2% to 3%	log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.22, OR 1.25; range across patients 0.22 to 0.22	122, 73	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to Arkansas (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds 0.14, OR 1.15; range across patients 0.14 to 0.14	122, 78	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C074	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	3% to 3%	log-odds -0.01, OR 0.99; range across patients -0.01 to -0.01	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0004, 5e-05], n=1000)	yes	slightly; on the training data only
C075	direction: state: West Virginia to Florida (through this part) *shape: categorical; highest to lowest: Washington, New Jersey, Connecticut, Louisiana, Tennessee, Massachusetts …*	3% to 3%	log-odds -0.0006, OR 1.00; range across patients -0.0006 to -0.0006	122, 78	train; test disagrees (AUC gain -0.008 [-0.022, 0.005], n=1000)	yes	slightly; on the training data only
C076	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		log-odds 1e-15, OR n/a	4000	train	yes	
C077	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.19, OR 1.21; range across patients 0.19 to 0.19	4000	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	slightly
C078	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 18% (typical patient with a fold)	fold 2.80 calls (IQR 2.80 to 2.80); past it log-odds 2.07, OR 7.90; range across patients 2.07 to 2.07	4000, 848	train; test agrees (AUC gain 0.111 [0.070, 0.152], n=1000)	yes	clearly
C079	structure: made of csc_moderate_chain, csc_high_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C080	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: hinge at 11 messages; 64% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	12% to 6% (typical patient this part acts on)	log-odds -0.26, OR 0.77; range across patients -1.08 to 0.00	4000	train; test agrees (AUC gain 0.006 [0.003, 0.009], n=1000)	yes	clearly
C081	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 221 minutes; 50% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 8%	log-odds 1.13, OR 3.10; range across patients 0.83 to 1.79	4000	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C082	direction: state: state other than Washington to Washington (through this part) *shape: categorical; highest to lowest: state other than Washington, Washington*	12% to 7% (typical patient this part acts on)	log-odds -0.21, OR 0.81; range across patients -0.84 to 0.00	3928, 72	train; test agrees (AUC gain 0.0004 [2e-05, 0.0008], n=1000)	yes	slightly
C083	direction: total daytime charge: 19.07 US dollars to 42.40 US dollars (through this part) *shape: hinge at 39.34 US dollars; 77% of patients same; fallback; risk saturates near 0%; one point with C090 (median fold 39.34 US dollars, typical patient 42.20 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	3% to 3%	log-odds 0.24, OR 1.27; range across patients 0.00 to 0.94	4000	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; on the training data only
C084	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge above 30.69 US dollars *shape: accelerating; 74% of patients same; fallback; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 6%	log-odds difference 0.54, OR 1.71	1997, 2003	train; test agrees (AUC gain 0.005 [-0.0004, 0.012], n=1000); group difference on test 0.53 [0.49, 0.57]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by state: Washington minus state other than Washington; stronger in state other than Washington *shape: hinge at 223 minutes; 51% of patients same; fallback; risk saturates near 0%*	Washington: 11% to 14%; state other than Washington: 3% to 8%	log-odds difference -0.75, OR 0.47	72, 3928	train; test agrees (AUC gain 0.0004 [2e-05, 0.0008], n=1000); group difference on test -0.80 [-0.92, -0.65]	yes	slightly
C086	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime charge: total daytime charge above 30.69 US dollars minus total daytime charge at or below 30.69 US dollars; stronger in total daytime charge at or below 30.69 US dollars *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime charge above 30.69 US dollars: 3% to 11%; total daytime charge at or below 30.69 US dollars: 3% to 11%	log-odds difference -0.53, OR 0.59	1997, 2003	train; test agrees (AUC gain 0.005 [-0.0004, 0.012], n=1000); group difference on test -0.55 [-0.61, -0.49]	yes	slightly
C087	interaction: effect of voice-mail messages (0 messages to 32 messages) by total daytime minutes: total daytime minutes above 180 minutes minus total daytime minutes at or below 180 minutes; stronger in total daytime minutes at or below 180 minutes *shape: steady; 100% of patients same; verified; risk saturates near 0%*	total daytime minutes above 180 minutes: 3% to 11%; total daytime minutes at or below 180 minutes: 3% to 11%	log-odds difference -0.53, OR 0.59	1997, 2003	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000); group difference on test -0.55 [-0.61, -0.49]	yes	slightly
C088	threshold: total daytime minutes: fold at about 221 minutes (starts) *shape: hinge at 221 minutes; 48% of patients same; fallback; risk saturates near 0%; one point with C088 (median fold 221 minutes, typical patient 223 minutes, 10th-90th percentile 186 minutes to 221 minutes, wide)*	3% to 19% (typical patient with a fold)	fold 221 minutes (IQR 213 to 221); past it log-odds 1.98, OR 7.25; range across patients 1.48 to 2.79	3995, 1099	train; test agrees (AUC gain 0.036 [0.015, 0.059], n=1000)	yes	clearly
C089	threshold: voice-mail messages: fold at about 11 messages (stops) for customers in New Jersey *shape: hinge at 11 messages; 74% of patients same; fallback; one point with C089 (median fold 11 messages, typical patient 11 messages, 10th-90th percentile 4 messages to 27 messages, wide)*	24% to 13% (typical patient with a fold (customers in New Jersey))	fold 10.8 messages (IQR 6.77 to 14.9); past it log-odds -0.93, OR 0.40; range across patients -1.97 to -0.27	36, 26	train; test agrees (AUC gain 0.006 [0.003, 0.009], n=1000)	yes	clearly
C090	threshold: total daytime charge: fold at about 39.34 US dollars (starts) for customers in North Carolina *shape: hinge at 39.34 US dollars; 88% of patients same; fallback; one point with C090 (median fold 39.34 US dollars, typical patient 39.00 US dollars, 10th-90th percentile 26.14 US dollars to 52.75 US dollars, wide)*	4% to 5% (typical patient with a fold (customers in North Carolina))	fold 39.3 US dollars (IQR 34.2 to 45.8); past it log-odds 0.51, OR 1.67; range across patients 0.00 to 1.28	50, 17 (small n)	train; test unclear (AUC gain 0.005 [-0.0004, 0.012], n=1000)	yes	slightly; the model has little data on this; on the training data only
C091	subgroup: who this part acts on: 1099 of 4000 patients	3% to 12% (typical patient this part acts on (risk without the part, then with it))	share 0.27; log-odds 1.49, OR 4.42; range across patients 0.12 to 3.47	1099, 2901	train; test agrees (AUC gain 0.080 [0.048, 0.112], n=1000)	yes	clearly
C092	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C093	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 6% (typical patient with a fold)	fold 3.21 calls (IQR 3.21 to 3.21); past it log-odds 0.91, OR 2.49; range across patients 0.91 to 0.91	4000, 318	train; test agrees (AUC gain 0.023 [0.012, 0.035], n=1000)	yes	slightly
C094	subgroup: who this part acts on: 318 of 4000 patients	8% to 18% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 1.71, OR 5.51; range across patients 0.91 to 3.72	318, 3682	train; test agrees (AUC gain 0.023 [0.012, 0.035], n=1000)	yes	clearly
C095	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.0, OR n/a	4000	train	yes	
C096	direction: customer-service calls: 0	3% to 3%	log-odds 0.19, OR 1.21; range across patients	4000	train; test agrees (AUC gain 0.028	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	3% to 8% (typical patient with a fold)	fold 2.80 calls (IQR 2.80 to 2.80); past it log-odds 1.16, OR 3.18; range across patients 1.16 to 1.16	4000, 848	train; test agrees (AUC gain 0.028 [0.014, 0.042], n=1000)	yes	clearly
C098	subgroup: who this part acts on: 848 of 4000 patients	3% to 4% (typical patient this part acts on (risk without the part, then with it))	share 0.21; log-odds 0.76, OR 2.13; range across patients 0.19 to 2.12	848, 3152	train; test agrees (AUC gain 0.028 [0.014, 0.042], n=1000)	yes	slightly
C099	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C100	direction: state: state other than Indiana, Rhode Island to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 22%	log-odds 2.36, OR 10.6; range across patients 2.36 to 2.36	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	on the training data only
C101	direction: state: state other than Indiana, Rhode Island to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Indiana, state other than Indiana, Rhode Island*	3% to 3%	log-odds 0.27, OR 1.31; range across patients 0.27 to 0.27	3836, 82	train; test unclear (AUC gain 0.005 [-0.002, 0.016], n=1000)	yes	slightly; on the training data only
C102	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C103	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3%	log-odds 0.20, OR 1.22; range across patients 0.20 to 0.20	4000	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C104	threshold: total evening minutes: fold at about 217 minutes (starts) *shape: hinge at 217 minutes; 100% of patients same; verified; risk saturates near 0%; one point with C104 (median fold 217 minutes, typical patient 218 minutes, 10th-90th percentile 217 minutes to 217 minutes)*	3% to 3% (typical patient with a fold)	fold 217 minutes (IQR 217 to 217); past it log-odds 0.28, OR 1.32; range across patients 0.28 to 0.28	4000, 1467	train; test unclear (AUC gain 0.003 [-8e-05, 0.005], n=1000)	yes	slightly; on the training data only
C105	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C106	direction: state: state other than Louisiana to Louisiana (through this part) *shape: categorical; highest to lowest: Louisiana, state other than Louisiana*	3% to 3%	log-odds 0.04, OR 1.04; range across patients 0.04 to 0.05	3935, 65	train; test disagrees (AUC gain -6e-05 [-0.0002, 5e-05], n=1000)	yes	slightly; on the training data only
C107	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3%	log-odds 0.004, OR 1.00; range across patients 0.003 to 0.003	4000	train; test unclear (AUC gain 2e-05 [-9e-05, 0.0001], n=1000)	yes	slightly; on the training data only
C108	interaction: effect of voice-mail messages (0 messages to 32 messages) by state: Louisiana minus state other than Louisiana; stronger in Louisiana *shape: steady; 78% of patients same; fallback*	Louisiana: 5% to 19%; state other than Louisiana: 3% to 11%	log-odds difference 0.14, OR 1.15	65, 3935	train; test unclear (AUC gain -6e-05 [-0.0002, 5e-05], n=1000); group difference on test 0.11 [-0.15, 0.29]	yes	on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	threshold: voice-mail messages: fold at about 22 messages (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	3% to 3% (typical patient with a fold)	fold 21.0 messages (IQR 21.0 to 21.0); past it log-odds 0.005, OR 1.00; range across patients 0.005 to 0.005	3935, 876	train; test unclear (AUC gain 2e-05 [-9e-05, 0.0001], n=1000)	no	slightly; on the training data only
C110	subgroup: who this part acts on: 941 of 4000 patients	2% to 2% (typical patient this part acts on (risk without the part, then with it))	share 0.24; log-odds 0.006, OR 1.01; range across patients 0.0009 to 0.007	941, 3059	train; test disagrees (AUC gain -5e-05 [-0.0002, 9e-05], n=1000)	no	slightly; on the training data only
C111	null: this part barely changes predictions (switched off)		max change 0.01 (max |change in predicted risk|)	4000	train	yes	
C112	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C113	boundary: boundary		log-odds 0.00, OR n/a	4000	train	no	

Not supported by the model.

- C109: threshold claim about number of voice-mail messages in part ann_84: not supported by the model (the fold changes the curve by only 0.007 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C110: subgroup claim about state and number of voice-mail messages in part ann_84: not supported by the model (its effect is negligible).
- C113: boundary claim about total daytime minutes and number of voice-mail messages in part ann_82: not supported by the model (the boundary rule matches 0.0 of training rows (needs 0.99)).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (37):** `-1` (state:0), `-14` (state:20), `-15` (state:21), `-16` (state:22), `-17` (state:23), `-18` (state:24), `-19` (state:25), `-2` (state:1), `-20` (state:26), `-21` (state:27), `-23` (state:29), `-25` (state:30), `-27` (state:32), `-28` (state:33), `-29` (state:34), `-30` (state:35), `-35` (state:4), `-36` (state:40), `-39` (state:43), `-40` (state:44), `-44` (state:48), `-46` (state:5), `-47` (state:50), `-49` (state:7), `-5` (state:12), `-50` (state:8), `-53` (areaCode:408), `-55` (areaCode:510), `-6` (state:13), `-63` (totalEveCalls), `-65` (totalNightMinutes), `-66` (totalNightCalls), `-67` (totalNightCharge), `-69` (totalIntlCalls), `-7` (state:14), `-70` (totalIntlCharge), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, . . .) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)

  ▷ A2: linear_backbone (leaf, section 1.6.2)

- ▷ A3: day_usage_hinges (leaf, section 1.6.3)
- ▷ A4: evening_hinge (leaf, section 1.6.4)
- ▷ A5: vmail_state18_unit (leaf, section 1.6.5)
- ▷ A6: state_offsets (leaf, section 1.6.6)
- ▷ A7: service_calls (composition, section 1.6.7)
  - ▷ A8: csc_moderate_chain (leaf, section 1.6.8)
  - ▷ A9: csc_high_hinge (leaf, section 1.6.9)

1.4 Network figures

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 4.44×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`, `-71_d`.
- #72: split `-58` into `-58_a`, `-58_b`, `-58_c`.
- #73: split `-43` into `-43_a`, `-43_b`.
- #74: split `-34` into `-34_a`, `-34_b`.
- #75: split `-11` into `-11_a`, `-11_b`.
- #76: consolidate `-71_a`, `-71_c`, `-71_d`.
- #77: identity node `identity_lin` before `0` (29 connection(s)).
- #78: identity node `identity_svc` before `0` (3 connection(s)).

1.6 Annotations

1.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6), A7 (section 1.6.7).

- **Entry nodes:** `-70` (totalIntlCharge), `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-64` (totalEveCharge), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-60` (totalDayCalls), `-59` (totalDayMinutes), `-57` (voiceMailPlan), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-42` (state:46), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-37` (state:41), `-36` (state:40), `-35` (state:4), `-33` (state:38), `-32` (state:37), `-31` (state:36), `-30` (state:35), `-29` (state:34), `-28` (state:33), `-27` (state:32), `-26` (state:31), `-25` (state:30), `-24` (state:3), `-23` (state:29), `-22` (state:28), `-21` (state:27), `-20` (state:26), `-19` (state:25), `-18` (state:24), `-17` (state:23), `-16` (state:22), `-15` (state:21), `-14` (state:20), `-13` (state:2), `-12` (state:19), `-10` (state:17), `-9` (state:16), `-8` (state:15), `-7` (state:14), … (17 more; see the JSON record).
- **Exit nodes:** `0`.
- **Internal nodes:** none.
- **Edges (weight):** `1295`→`0` (-1.01), `3118`→`0` (1.53), `6089`→`0` (0.00514), `7872`→`0` (-1.21), `9954`→`0` (0.0405), `10131`→`0` (1.91), `15524`→`0` (0.201), `identity_lin`→`0` (1), `identity_svc`→`0` (1).

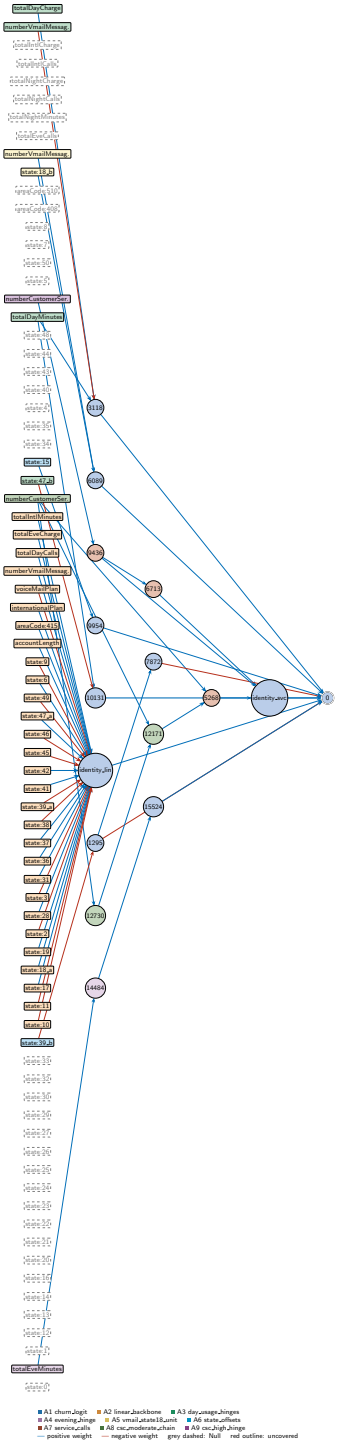


Figure 1: churn100-s4-c2: full view of the explained model state.

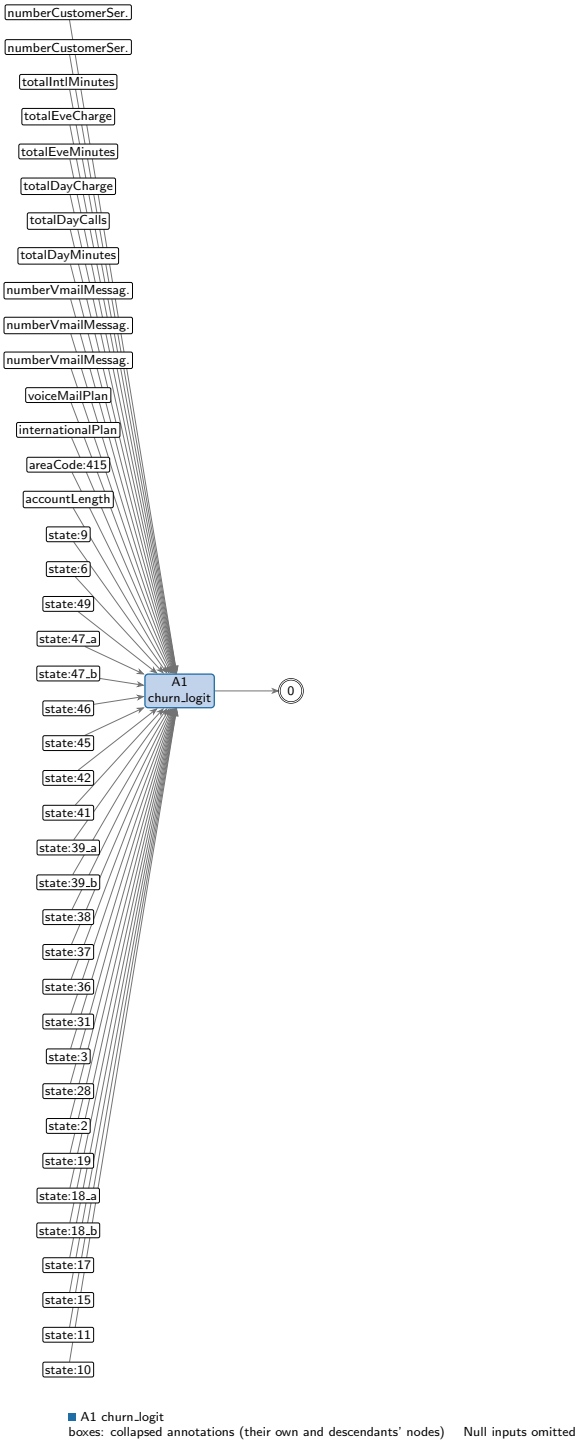


Figure 2: churn100-s4-c2: collapsed view at hierarchy level 0 (fully recomposed).

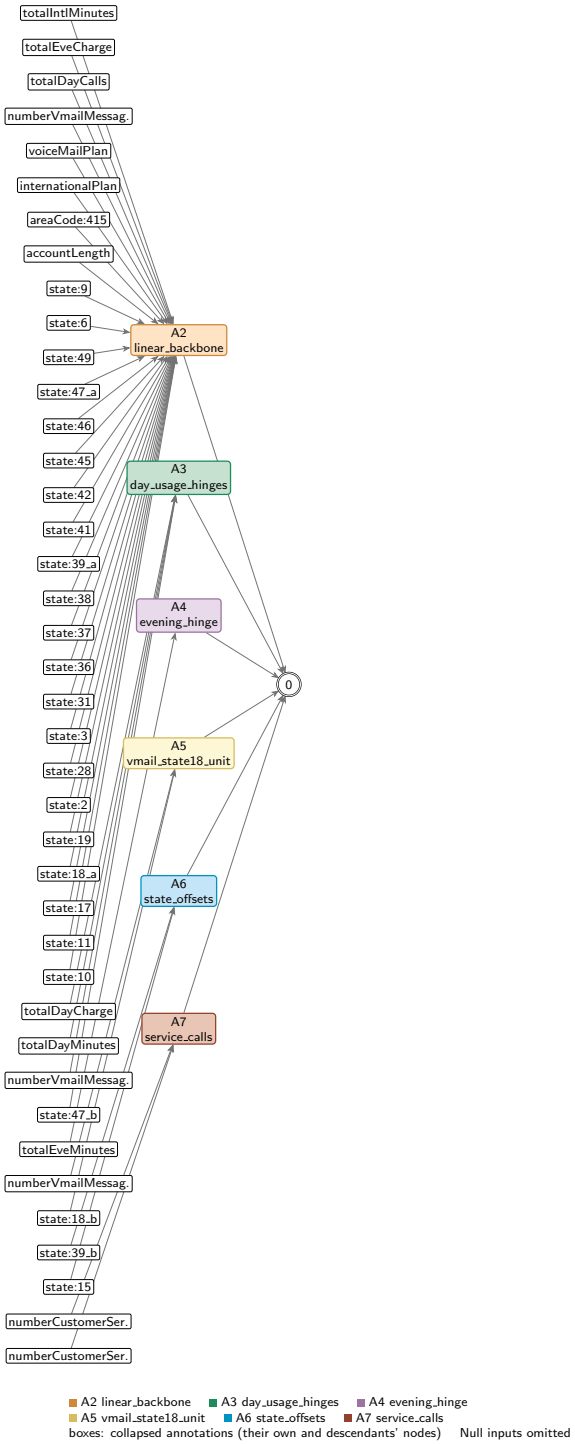


Figure 3: churn100-s4-c2: collapsed view at hierarchy level 1.

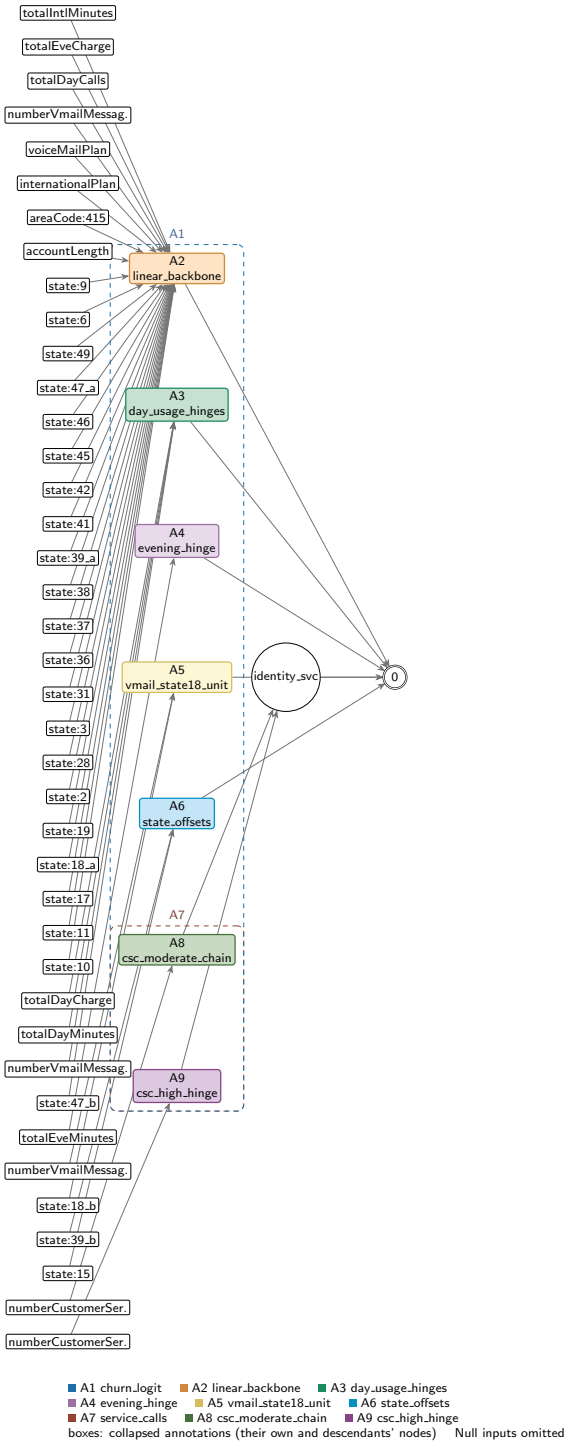


Figure 4: churn100-s4-c2: collapsed view at hierarchy level 2.

- **With its descendants:** 93 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 92 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/ calls, evening calls, area codes 408/510 and 29 state indicators) and have zero effect. Train AUC 0.8908, accuracy 0.888. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8333 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_ usage_hinges alone gives 0.8707, service_calls alone 0.8659, both 0.8907; evening, state and voicemail units restore nothing. So the gain over linear (+0.058) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.75 logit (73 rows, empirical -1.35); state:46 -1.94 (80 rows, -0.91); state:45 -0.94 (91, -0.85); state:47 +0.90 mean (72 rows, empirical +0.92; direct +1.23, but it switches off 10131 so heavy day users lose up to 2.75); state:31 +0.83 (90, +0.67); state:18 +0.41 (65 rows, empirical -0.48: opposite sign); state:39 -0.25 net (82, -0.93: direct -2.61 almost cancelled by the state units).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -1.3 + 1.0\,\mathrm{linear_backbone}\big(x_{state:17}, x_{state:18_a},\\
&\quad x_{state:19}, x_{state:2}, x_{state:28}, x_{state:3}, x_{state:31}, x_{state:10},\\
&\quad x_{state:36}, x_{state:37}, x_{state:38}, x_{state:39_a}, x_{state:41}, x_{state:42},\\
&\quad x_{state:11}, x_{state:45}, x_{state:46}, x_{state:47_a}, x_{state:49}, x_{state:6},\\
&\quad x_{state:9}, x_{accountLength}, x_{areaCode:415}, x_{internationalPlan}, x_{voiceMailPlan},\\
&\quad x_{numberVmailMessages_a}, x_{totalDayCalls}, x_{totalEveCharge}, x_{totalIntlMinutes}\big)\\
&\quad + 1.0\,\mathrm{service_calls}\big(x_{numberCustomerServiceCalls_acd},\\
&\quad x_{numberCustomerServiceCalls_b}\big)\\
&\quad + 0.201\,\mathrm{evening_hinge}\big(x_{totalEveMinutes}\big)\\
&\quad + 1.53\,\mathrm{day_usage_hinges_0}\big(x_{state:47_b}, x_{numberVmailMessages_b},\\
&\quad x_{totalDayCharge}, x_{totalDayMinutes}\big)\\
&\quad + 0.00514\,\mathrm{vmail_state18_unit}\big(x_{state:18_b},\\
&\quad x_{numberVmailMessages_c}\big) + 0.0405\,\mathrm{state_offsets_2}\big(x_{state:15},\\
&\quad x_{state:39_b}\big) + 1.91\,\mathrm{day_usage_hinges_1}\big(x_{state:47_b},\\
&\quad x_{numberVmailMessages_b}, x_{totalDayCharge}, x_{totalDayMinutes}\big)\\
&\quad - 1.01\,\mathrm{state_offsets_0}\big(x_{state:15}, x_{state:39_b}\big)\\
&\quad - 1.21\,\mathrm{state_offsets_1}\big(x_{state:15}, x_{state:39_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -1.3 + 0.00719x_{areaCode:415} + 0.201\,\mathrm{relu}\big(0.714\\
&\quad + 1.02\,\mathrm{relu}\big(-0.33 + 1.01x_{totalEveMinutes}\big)\big)\\
&\quad + 0.106x_{accountLength} + 0.0569x_{state:6} + 0.163x_{state:47_a}\\
&\quad + 1.53\,\mathrm{relu}\big(-0.856 + 0.382x_{totalDayCharge} + 0.547x_{totalDayMinutes}\\
&\quad - 0.65x_{numberVmailMessages_b}\big) + 1.76\,\mathrm{relu}\big(-0.582\\
&\quad + 0.462x_{numberCustomerServiceCalls_b}\big) + 0.133x_{totalDayCalls}\\
&\quad + 0.00514\,\mathrm{relu}\big(-0.668 + 1.14x_{state:18_b}\\
&\quad + 0.828x_{numberVmailMessages_c}\big) + 0.337x_{totalIntlMinutes} + 0.044x_{state:42}\\
&\quad + 1.66\,\mathrm{relu}\big(-0.557 + 0.358\,\mathrm{relu}\big(-0.443
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.719\,\mathrm{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\
&+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\
&+ 0.254 x_{numberCustomerServiceCalls_acd}\big) + 0.007 x_{state:36} + 0.00781 x_{state:41} \\
&+ 0.621 x_{numberVmailMessages_a} + 0.0397 x_{state:37} \\
&+ 0.0405\,\mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big) + 0.35 x_{totalEveCharge} \\
&+ 0.0453 x_{state:19} + 0.047 x_{state:18_a} + 1.54\,\mathrm{relu}\big(-0.127 \\
&+ 1.43\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \\
&+ 0.826 x_{internationalPlan} + 0.000897 x_{state:28} + 1.91\,\mathrm{relu}\big(-0.721 \\
&+ 0.837 x_{totalDayMinutes} - 0.672 x_{state:47_b}\big) + 0.0307 x_{state:17} \\
&+ 0.124 x_{state:31} - 0.37 x_{state:39_a} - 0.0355 x_{state:9} - 0.0158 x_{state:2} \\
&- 0.00422 x_{state:3} - 0.271 x_{state:46} - 1.12 x_{voiceMailPlan} - 0.141 x_{state:45} \\
&- 0.044 x_{state:49} - 1.01\,\mathrm{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
&- 0.0737 x_{state:38} - 1.21\,\mathrm{relu}\big(0.0138 \\
&+ 1.02\,\mathrm{relu}\big(0.868 - 1.27 x_{state:39_b}\big)\big) - 0.0797 x_{state:10} \\
&- 0.368 x_{state:11}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00219, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.30 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.53*3118 + 1.91*10131 (day_usage_hinges) + 0.20*15524 (evening_hinge) - 1.01*1295 - 1.21*7872 + 0.04*9954 (state_offsets) + 0.005*6089, purely additive in the logit; the sigmoid is the only coupling. 37 inputs are unconnected (international charge and calls, night minutes/charge/calls, evening calls, ar

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-10` (state:17), `-11_a` (state:18_a), `-12` (state:19), `-13` (state:2), `-22` (state:28), `-24` (state:3), `-26` (state:31), `-3` (state:10), `-31` (state:36), `-32` (state:37), `-33` (state:38), `-34_a` (state:39_a), `-37` (state:41), `-38` (state:42), `-4` (state:11), `-41` (state:45), `-42` (state:46), `-43_a` (state:47_a), `-45` (state:49), `-48` (state:6), `-51` (state:9), `-52` (accountLength), `-54` (areaCode:415), `-56` (internationalPlan), `-57` (voiceMailPlan), `-58_a` (numberVmailMessages_a), `-60` (totalDayCalls), `-64` (totalEveCharge), `-68` (totalIntlMinutes).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-3`→`identity_lin` (-0.0797), `-4`→`identity_lin` (-0.368), `-10`→`identity_lin` (0.0307), `-12`→`identity_lin` (0.0453), `-13`→`identity_lin` (-0.0158), `-22`→`identity_lin` (8.97×10^{-4}), `-24`→`identity_lin` (-0.00422), `-26`→`identity_lin` (0.124), `-31`→`identity_lin` (0.007), `-32`→`identity_lin` (0.0397), `-33`→`identity_lin` (-0.0737), `-37`→`identity_lin` (0.00781), `-38`→`identity_lin` (0.044), `-41`→`identity_lin` (-0.141), `-42`→`identity_lin` (-0.271), `-45`→`identity_lin` (-0.044), `-48`→`identity_lin` (0.0569), `-51`→`identity_lin` (-0.0355), `-52`→`identity_lin` (0.106), `-54`→`identity_lin` (0.00719), `-56`→`identity_lin` (0.826), `-57`→`identity_lin` (-1.12), `-60`→`identity_lin` (0.133), `-64`→`identity_lin` (0.35),

-68→identity_lin (0.337), -58_a→identity_lin (0.621), -43_a→identity_lin (0.163), -34_a→identity_lin (-0.37), -11_a→identity_lin (0.047).

- **Covers** (coverage, own and as part of the hierarchy): 29 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state offsets: state:11 -0.37 (-2.75 logit for its 73 customers), state:39 -0.37 (-2.61, but the state_offsets units give back +2.36), state:46 -0.27 (-1.94), state:47 +0.16 (+1.23), state:31 +0.12 (+0.83), state:45 -0.14 (-0.94), then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.12, voicemail messages 0.62, international plan 0.83).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.00719x_{areaCode:415} + 0.106x_{accountLength} + 0.0569x_{state:6} \\
& + 0.163x_{state:47_a} + 0.133x_{totalDayCalls} + 0.337x_{totalIntlMinutes} \\
& + 0.044x_{state:42} + 0.007x_{state:36} + 0.00781x_{state:41} \\
& + 0.621x_{numberVmailMessages_a} + 0.0397x_{state:37} + 0.35x_{totalEveCharge} \\
& + 0.0453x_{state:19} + 0.047x_{state:18_a} + 0.826x_{internationalPlan} \\
& + 0.000897x_{state:28} + 0.0307x_{state:17} + 0.124x_{state:31} - 0.37x_{state:39_a} \\
& - 0.0355x_{state:9} - 0.0158x_{state:2} - 0.00422x_{state:3} - 0.271x_{state:46} \\
& - 1.12x_{voiceMailPlan} - 0.141x_{state:45} - 0.044x_{state:49} - 0.0737x_{state:38} \\
& - 0.0797x_{state:10} - 0.368x_{state:11}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00544, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 29 inputs: voicemail plan -1.12 per SD (-2.54 logit for having a plan, mostly offset for heavy message users by voicemail messages +0.62 per SD), international plan +0.83 (+2.82 logit for having one), evening charge +0.35, international minutes +0.34, day calls +0.13, account length +0.11, and state o

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: day_usage_hinges

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** -43_b (state:47_b), -58_b (numberVmailMessages_b), -61 (totalDayCharge), -59 (totalDayMinutes).
- **Exit nodes:** 3118, 10131.
- **Internal nodes:** none.
- **Edges (weight):** -59→3118 (0.546), -61→3118 (0.382), -59→10131 (0.837), -58_b→3118 (-0.65), -43_b→10131 (-0.672).

- **Covers** (coverage, own and as part of the hierarchy): 4 candidate nodes.

Hypothesis. 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customers with no voicemail messages (its kink moves up about 25 minutes per 10 messages: 267 at 20, 292 at 30), so it is on for only 2% of message users; 10131 switches on at about 221 day minutes for everyone except state:47 (off for all 72 state:47 customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.62 at 220-240, +1.61 at 240-260, +2.59 at 260-280, +4.26 above 280; churn by band 0.06 (200-220), 0.14, 0.26, 0.46, 0.67 vs model 0.08, 0.15, 0.28, 0.43, 0.68 (well calibrated). Sensitivity: 3118 vmail messages 0.15, day minutes 0.12, day charge 0.09; 10131 day minutes 0.18, state:47 0.14. Mean ablation AUC 0.8107 (the largest single drop); linear replacement 0.8657 (R2 0.46/0.45); restored alone onto all-linear 0.8707 (all-linear 0.8333).

Closed form:

$$
\begin{aligned}
y_0 = \mathrm{relu}\big(&-0.856 + 0.382 x_{totalDayCharge} + 0.547 x_{totalDayMinutes} \\
&- 0.65 x_{numberVmailMessages_b}\big) \\
y_1 = \mathrm{relu}\big(&-0.721 + 0.837 x_{totalDayMinutes} \\
&- 0.672 x_{state:47_b}\big)
\end{aligned}
$$

where y_0 is node `3118`; y_1 is node `10131`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 8.05×10^{-4}, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 3118 = relu(-0.86 + 0.55 dayMin + 0.38 dayCharge - 0.65 vmailMessages), logit weight +1.53; 10131 = relu(-0.72 + 0.84 dayMin - 0.67 state:47), logit weight +1.91. Day minutes and day charge are collinear (r = 1.0000), so 3118 is one hinge in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply. 3118 switches on at about 209 day minutes for customer

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.4 A4: evening_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-62` (totalEveMinutes).
- **Exit nodes:** `15524`.
- **Internal nodes:** `14484`.
- **Edges (weight):** `-62`→`14484` (1.01), `14484`→`15524` (1.02).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation AUC 0.8907, linear replacement 0.8907 (R2 0.62): negligible for ranking.

Closed form:

$$y_0 = \mathrm{relu}\big(0.714 + 1.02\,\mathrm{relu}\big(-0.33 + 1.01 x_{totalEveMinutes}\big)\big)$$

where y_0 is node `15524`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0165, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 14484 = relu(-0.33 + 1.01 eveMinutes), 15524 = relu(0.71 + 1.02*14484) (always active); logit weight +0.20. Hypothesis: a small hinge in evening minutes, constant (+0.14 logit) below about 217 evening minutes and rising steadily above (+0.21 at 220-250, +0.36 at 250-300, +0.56 above 300); it adds to the direct evening-charge term (+0.35 per SD). Evidence: line plot correlation 0.79; mean ablation

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: vmail_state18_unit

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-11_b` (state:18_b), `-58_c` (numberVmailMessages_c).
- **Exit nodes:** `6089`.
- **Internal nodes:** none.
- **Edges (weight):** `-58_c`→`6089` (0.828), `-11_b`→`6089` (1.14).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.668 + 1.14 x_{state:18_b} + 0.828 x_{numberVmailMessages_c}\big)$$

where y_0 is node `6089`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.02, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 6089 = relu(-0.67 + 1.14 state:18 + 0.83 vmailMessages), logit weight +0.005. On for customers with more than about 21 voicemail messages and for all 65 state:18 customers; logit contribution at most +0.05. Hypothesis: negligible. Evidence: mean ablation and linear replacement both leave train AUC at 0.8908.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.6 A6: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-8` (state:15), `-34_b` (state:39_b).
- **Exit nodes:** `1295`, `7872`, `9954`.
- **Internal nodes:** none.
- **Edges (weight):** `-8`→`9954` (1.01), `1295`→`7872` (1.02), `-34_b`→`1295` (-1.27).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling state:39's direct -2.61 (net -0.25, against an empirical log-odds offset of -0.93 for its 82 customers); 9954 adds +0.27 for state:15. Evidence: sensitivity 1.24/1.27 for state:39, 0.02 for state:15; linear replacement AUC 0.8908 (unchanged); mean ablation 0.8898.

Closed form:

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(0.868 - 1.27 x_{state:39_b}\big) \\
y_1 &= \operatorname{relu}\big(0.0138 + 1.02 \operatorname{relu}\big(0.868 \\
&\qquad - 1.27 x_{state:39_b}\big)\big) \\
y_2 &= \operatorname{relu}\big(-0.38 + 1.01 x_{state:15}\big)
\end{aligned}
$$

where y_0 is node `1295`; y_1 is node `7872`; y_2 is node `9954`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00698, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 1295 = relu(0.87 - 1.27 state:39) -> 7872 = relu(0.01 + 1.02*1295) (logit weights -1.01, -1.21); 9954 = relu(-0.38 + 1.01 state:15) (weight +0.04). Each depends only on one binary state indicator, so each is exactly an offset (least-squares R2 = 1.0). Hypothesis: the pair is a constant -2.33 logit for everyone except state:39 customers, for whom it is switched off (+2.36 logit), almost cancelling

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.7 A7: service_calls

Kind: Composition annotation (depth 1); parent A1 (section 1.6.1); children A8 (section 1.6.8), A9 (section 1.6.9).

- **Entry nodes:** `-71_acd` (numberCustomerServiceCalls_acd), `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `5268`→`identity_svc` (1.66), `9436`→`identity_svc` (1.76), `6713`→`identity_svc` (1.54).

- **With its descendants:** 8 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 7 candidate nodes.

Hypothesis. Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at 4, +4.44 at 5, +6.81 at 6, +9.18 at 7). Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.118/0.108/0.105/0.115/0.327/0.749/0.956/0.986: the threshold is right, but the model under-predicts at 4 calls and is over-confident from 5 calls (116 customers). Mean-ablating both children gives AUC 0.7976, linear replacement 0.8711; restoring service_calls alone onto the all-linear model gives 0.8659 (all-linear 0.8333).

Closed form (children as named functions):

$$\begin{aligned} y_0 = {} & 1.76\,\mathrm{csc_high_hinge_0}\big(x_{numberCustomerServiceCalls_b}\big) \\ & + 1.66\,\mathrm{csc_moderate_chain}\big(x_{numberCustomerServiceCalls_acd}\big) \\ & + 1.54\,\mathrm{csc_high_hinge_1}\big(x_{numberCustomerServiceCalls_b}\big) \end{aligned}$$

Expanded over the children:

$$\begin{aligned} y_0 = {} & 1.76\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\ & + 1.66\,\mathrm{relu}\big(-0.557 + 0.358\,\mathrm{relu}\big(-0.443 \\ & + 0.719\,\mathrm{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ & + 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ & + 0.254 x_{numberCustomerServiceCalls_acd}\big) + 1.54\,\mathrm{relu}\big(-0.127 \\ & + 1.43\,\mathrm{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \end{aligned}$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0106, and the exact function by at most 1.78×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_moderate_chain and csc_high_hinge: their exits are summed by identity_svc = 1.66*5268 + 1.76*9436 + 1.54*6713, which enters the logit with weight 1; they read the same input (two split copies of customer-service calls). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.19 logit at 3 calls, then about +2.4 logit per extra call (+2.07 at

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1.6.8 A8: csc_moderate_chain

Kind: Leaf annotation (depth 2); parent A7 (section 1.6.7).

- **Entry nodes:** `-71_acd` (numberCustomerServiceCalls_acd).
- **Exit nodes:** `5268`.
- **Internal nodes:** `12730`, `12171`.
- **Edges (weight):** `12171`→`5268` (0.358), `12730`→`12171` (0.719), `-71_acd`→`5268` (0.254), `-71_acd`→`12171` (0.845), `-71_acd`→`12730` (0.801).

- **Covers** (coverage, own and as part of the hierarchy): 3 candidate nodes.

Hypothesis. 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): line plot of 5268 against calls rises from 0 to 3.6 (correlation 0.74); 5268 active on 21% of customers (all with 3+ calls). Mean ablation AUC 0.8733 (full 0.8908); linear replacement 0.8852 (R2 0.55); restored alone onto the all-linearised model 0.8565 (all-linear 0.8333).

Closed form:

$$\begin{aligned} y_0 = \operatorname{relu}\big(&-0.557 + 0.358\operatorname{relu}\big(-0.443 \\ &+ 0.719\operatorname{relu}\big(-0.0259 + 0.801 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.845 x_{numberCustomerServiceCalls_acd}\big) \\ &+ 0.254 x_{numberCustomerServiceCalls_acd}\big) \end{aligned}$$

where y_0 is node `5268`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00332, and the exact function by at most 1.11×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 12730 = relu(-0.03 + 0.80 csc), 12171 = relu(-0.44 + 0.84 csc + 0.72*12730), 5268 = relu(-0.56 + 0.25 csc + 0.36*12171); logit weight +1.66 (via identity_svc). Hypothesis: a convex hinge on customer-service calls that is zero for 0-2 calls, just switches on at 3 calls (mean logit +0.19) and then adds about +1.0 logit per extra call (+1.16 at 4, +2.12 at 5, +3.09 at 6). Evidence (train, n=4000): li

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1.6.9 A9: csc_high_hinge

Kind: Leaf annotation (depth 2); parent A7 (section 1.6.7).

- **Entry nodes:** `-71_b` (numberCustomerServiceCalls_b).
- **Exit nodes:** `9436`, `6713`.
- **Internal nodes:** none.
- **Edges (weight):** `9436`→`6713` (1.43), `-71_b`→`9436` (0.462).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot correlation 0.66; mean ablation AUC 0.8761; linear replacement 0.8857 (R2 0.44/0.41); restored alone onto all-linear 0.8571.

Closed form:

$$\begin{aligned} y_0 &= \operatorname{relu}\big(-0.582 + 0.462 x_{numberCustomerServiceCalls_b}\big) \\ y_1 &= \operatorname{relu}\big(-0.127 + 1.43\operatorname{relu}\big(-0.582 \\ &\qquad + 0.462 x_{numberCustomerServiceCalls_b}\big)\big) \end{aligned}$$

where y_0 is node `9436`; y_1 is node `6713`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0119, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 9436 = relu(-0.58 + 0.46 csc), 6713 = relu(-0.13 + 1.43*9436); logit weights +1.76 and +1.54 (via identity_svc). Hypothesis: a second hinge on customer-service calls, off for 0-3 calls and switching on from 4 calls, adding about +1.4 logit per extra call (+0.91 at 4, +2.32 at 5, +3.72 at 6). Active on exactly the 8% of customers with 4+ calls (318 rows; empirical churn jumps from 0.11 at 3 calls t

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1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft